

NER (Named Entity Recognition) Tags across Packages

BERT, spaCy, Stanford CoreNLP, Stanza

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NER Approaches in the NLP Suite

BERT

For Named Entity Recognition the NLP Suite uses the RoBERTa model `djagatiya/ner-roberta-base-ontonotesv5-englishv4`, fine-tuned on the OntoNotes 5 corpus. This model processes the English language only.

Because it is trained on OntoNotes, BERT returns the same 18 entity types as spaCy and Stanza (English) — see the spaCy table below for a plain-language description of each: PERSON, NORP, ORG, GPE, LOC, FAC, PRODUCT, EVENT, WORK_OF_ART, LAW, LANGUAGE, DATE, TIME, PERCENT, MONEY, QUANTITY, ORDINAL, CARDINAL

(An earlier BERT model used a smaller, coarser tag set — geo, org, per, gpe, tim, art, eve, nat. It has been replaced; those tags no longer apply.)

spaCy

spaCy includes a list of 18 entity types:

Entity Type	Type Description
CARDINAL	A plain number that isn't another type - e.g., 'three', '1,000'.
DATE	A date or time period - e.g., 'June 2026', 'last summer'.
EVENT	A named event - wars, battles, hurricanes, sports events (e.g., 'World War II').
FAC	FAC = 'facility'. A man-made structure - buildings, airports, highways, bridges.
GPE	GPE = 'geo-political entity'. A place that has a government - a country, a state/region, or a city (e.g., 'Italy', 'Atlanta').

Entity Type	Type Description
LANGUAGE	A named language - e.g., 'English', 'Italian'.
LAW	A named law or legal document - e.g., 'the Constitution'.
LOC	A natural place that is NOT a country or city - mountains, rivers, lakes, oceans, regions (e.g., 'the Alps', 'the Nile').
MONEY	An amount of money - e.g., '\$5', 'ten euros'.
NORP	NORP = Nationalities, Religious, or Political groups - e.g., 'Italians', 'Catholics', 'Democrats'.
ORDINAL	A position word - 'first', 'second', 'third'.
ORG	An organization - a company, agency, school, or team (e.g., 'Emory University').
PERCENT	A percentage - e.g., '20%'.
PERSON	A person, real or fictional - e.g., 'Harry Potter'.
PRODUCT	An object or thing - vehicles, foods, gadgets (not services).
QUANTITY	A measurement - weight or distance (e.g., '5 km', 'two pounds').
TIME	A time shorter than a day - e.g., '3 p.m.', 'two hours'.
WORK_OF_ART	The title of a creative work - a book, song, or painting.

Stanford CoreNLP

PERSON
 ORGANIZATION
 MISC
 MONEY
 NUMBER
 ORDINAL
 PERCENT
 DATE
 TIME
 DURATION
 SETE
 MAIL
 URL
 CITY
 STATE_OR_PROVINCE
 COUNTRY
 LOCATION
 NATIONALITY
 RELIGION
 TITLE
 IDEOLOGY

CRIMINAL_CHARGE
CAUSE_OF_DEATH

Stanza

https://stanfordnlp.github.io/stanza/ner_models.html

Stanza's English model uses the same 18 OntoNotes entity types as spaCy - see the spaCy table above for a plain-language description of each (PERSON, GPE, LOC, FAC, NORP, ORG, DATE, and so on).

Important: Stanza tags cities, states, and countries all as GPE. Unlike Stanford CoreNLP, it does NOT separate CITY, STATE_OR_PROVINCE, and COUNTRY.

Stanza's other (non-English) languages use a smaller set of tags. The most common ones mean:

PER / PERS / PERSON - a person.

LOC / LOCATION - a place (city, country, region, or natural feature). In languages such as Italian, French, German and Spanish, every place is tagged LOC.

ORG / ORGANIZATION - an organization (company, agency, institution).

MISC / MISCELLANEOUS - a named thing that does not fit the other groups.

EVT / EVENT - a named event. PRO / PRODUCT - a product or object. DATE / TIME - a date / a time.

NE, NUM, PNAME, RACE (Burmese) - named entity, number, person name, and ethnic/national group.

The full per-language tag lists follow:

```
"fr": [  
  "LOC",  
  "MISC",  
  "ORG",  
  "PER"  
],  
"en": [  
  "CARDINAL",  
  "DATE",  
  "EVENT",  
  "FAC",  
  "GPE",  
  "LANGUAGE",  
  "LAW",  
  "LOC",  
  "MONEY",  
  "NORP",  
  "ORDINAL",  
  "ORG",  
  "PERCENT",
```

```
"PERSON",
"PRODUCT",
"QUANTITY",
"TIME",
"WORK_OF_ART"
],
"zh-hans": [
  "CARDINAL",
  "DATE",
  "EVENT",
  "FAC",
  "GPE",
  "LANGUAGE",
  "LAW",
  "LOC",
  "MONEY",
  "NORP",
  "ORDINAL",
  "ORG",
  "PERCENT",
  "PERSON",
  "PRODUCT",
  "QUANTITY",
  "TIME",
  "WORK_OF_ART"
],
"ru": [
  "LOC",
  "MISC",
  "ORG",
  "PER"
],
"uk": [
  "LOC",
  "MISC",
  "ORG",
  "PERS"
],
"ar": [
  "LOC",
  "MISC",
  "ORG",
  "PER"
],
```

```
"hu": [  
  "LOC",  
  "MISC",  
  "ORG",  
  "PER"  
],  
"af": [  
  "LOC",  
  "MISC",  
  "ORG",  
  "PERS"  
],  
"bg": [  
  "EVT",  
  "LOC",  
  "ORG",  
  "PER",  
  "PRO"  
],  
"fi": [  
  "DATE",  
  "EVENT",  
  "LOC",  
  "ORG",  
  "PER",  
  "PRO"  
],  
"my": [  
  "LOC",  
  "NE",  
  "NUM",  
  "ORG",  
  "PNAME",  
  "RACE",  
  "TIME"  
],  
"it": [  
  "LOC",  
  "ORG",  
  "PER"  
],  
"de": [  
  "LOC",  
  "MISC",
```

```
    "ORG",
    "PER"
  ],
  "nl": [
    "LOC",
    "MISC",
    "ORG",
    "PER"
  ],
  "vi": [
    "LOCATION",
    "MISCELLANEOUS",
    "ORGANIZATION",
    "PERSON"
  ],
  "es": [
    "LOC",
    "MISC",
    "ORG",
    "PER"
  ]
]
```